

Electromagnetic Wave

Summer 2026

1 Electromagnetic wave in linear material

1.1 1D traveling wave

Consider a plane wave traveling in *the* z direction ,then we can express it in a simple sin / cos wave:

$$\mathbf{E} = \mathbf{E}_0 \cos(kz \pm \omega t + \delta_E) \quad (1)$$

Now,by considering the polar coordinate, the wave can be expressed as :

$$\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)} \quad (2)$$

Where:

$$\tilde{\mathbf{E}}_0 = \mathbf{E}_0 e^{i\delta_E}$$

Complex wave function = Complex amplitude $\times$ traveling wave

Note 1. how do we tell the direction the wave travels ?

First,we set up the rule that **we pick the** $e^{-i\omega t}$ **always**.Now,the traveling wave will become:

$$\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_0 e^{\pm i(kz - \omega t)}$$

choose $+$: travel toward $+z$

choose $-$: travel toward $-z$

Now,we talk about some physics quantity in electromagnetic wave. *Wave is a disturbance in a medium that progrates in fixed shape and velocity.And it only tansfer energy but no medium.*

k is wavenumber,it means the density in space and we define $k = \frac{2\pi}{\lambda}$. ω is angular frequency,it represents how fast the oscillation varies in time and we define $\omega = 2\pi f = \frac{2\pi}{T}$.

$$\begin{aligned} k &= \frac{2\pi}{\lambda} \implies \lambda = \frac{2\pi}{k} \\ \omega &= 2\pi f \implies f = \frac{\omega}{2\pi} \\ v &= v_p = f\lambda = \left(\frac{\omega}{2\pi}\right)\left(\frac{2\pi}{k}\right) = \frac{\omega}{k} \\ n &= \frac{c}{v} = \frac{c}{\frac{\omega}{k}} = c \frac{k}{\omega} \quad (\text{refractive index}) \end{aligned}$$

By(1),we can find that the the velocity of carrier wave(v_p) can be observed from the phase in $\cos(kz - \omega t)$ is constant.That is:

$$\begin{aligned} kz - \omega t &= \text{const.} \\ k \frac{dz}{dt} - \omega &= kv_p - \omega = 0 \\ \therefore v_p &= \frac{\omega}{k} \end{aligned}$$

Now, if we consider two plane waves progrates in $+z$ direction and two waves will interface with each other. ($k_1 \approx k_2, \omega_1 \approx \omega_2$) Therefore,

$$\mathbf{E} = \mathbf{E}_0 \cos(k_1 z - \omega_1 t) + \mathbf{E}_0 \cos(k_2 z - \omega_2 t)$$

Using the trigonometric identity $\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$:

$$\therefore \mathbf{E} = 2\mathbf{E}_0 \underbrace{\cos \left(\frac{\Delta k}{2} z - \frac{\Delta \omega}{2} t \right)}_{\text{envelope wave}} \underbrace{\cos (\bar{k} z - \bar{\omega} t)}_{\text{carrier wave}}$$

From the derivation above, we find that when two wave superpose, they will become an wave packet with high frequency wave inside it. The envelope wave will have small frequency, wavenumber and varies very slowly. The wave packet modulates the carrier wave, which is the physical quantity that is actually responsible for transmitting energy and signals. The propagation speed is group velocity (v_g). We can express v_g as :

$$v_g = \left[\frac{\partial k}{\partial \omega} \right]^{-1}$$

In envelope wave, we let the phase to be the constant to get the group velocity.

$$\begin{aligned} kz - \omega t &= \text{const.} \\ \Delta k \frac{dz}{dt} - \Delta \omega &= \Delta k v_p - \Delta \omega = 0 \\ \therefore v_g &= \frac{\Delta \omega}{\Delta k} \end{aligned}$$

Therefore, by the definition of refractive index, we can get the group refractive index :

$$n_g \equiv \frac{c}{v_g} = c \left[\frac{\partial k}{\partial \omega} \right]$$

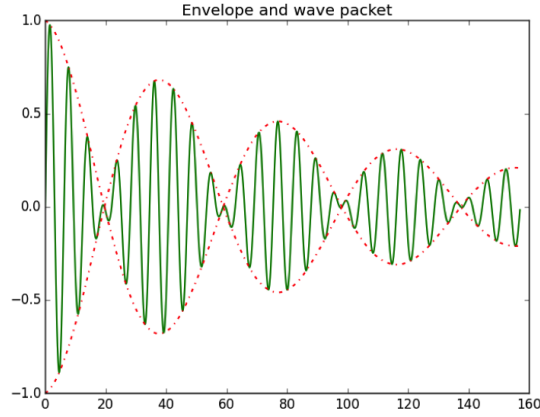


Figure 1: wavepacket

1.2 dispersive medium

First, we should discuss what is the dispersion. **Dispersion** means the propagation velocity of wave (phase velocity) will be affected by the operating frequency. That is $v_p = v_p(\omega)$. Therefore, the refractive index will be different in different frequency ($n = n(\lambda)$). In microscope, the different wave behavior is derived from the interaction between EM wave with bound charge ($\epsilon(\omega)$) and free charges ($\sigma(\omega)$).

$$\therefore v_p = \frac{1}{\sqrt{\mu\epsilon(\omega)}}$$

In fact, all wave parameters depending on ω :

$$\mu = \mu(\omega), \epsilon = \epsilon(\omega), \sigma = \sigma(\omega)$$

In the following chapter, we focus on EM wave in Perfect Linear Insulator (PLI) and in Perfect Linear Conductivity (PLC). We first list the physical parameters :

- μ (permeability)
- ϵ (permittivity)
- σ (conductivity)

Note 2. The subscript of parameter 0 means in the vacuum.

By the above parameters,we can substituting these into v and k :

$$\begin{aligned} k &= \frac{\omega}{v} = w\sqrt{\mu\epsilon} \\ v &= \frac{1}{\sqrt{\mu\epsilon}} \quad (\text{by wave equation}) \\ n &= \frac{c}{v} = \frac{\sqrt{\mu\epsilon}}{\sqrt{\mu_0\epsilon_0}} = \sqrt{\mu_r\epsilon_r} \end{aligned}$$

1.3 EM Wave in PLI

In the PLI,the parameters $(\mu, \epsilon, \sigma = 0)$.Insulators do not conduct electricity $(\sigma = 0)$.By *Ohm's law*, $\mathbf{J}_f = \sigma \mathbf{E}$.The free current density inside PLI is zero and thus $\rho_f = 0$ *under the condition that there is no charge pull into the materials*.

$$\mathbf{J}_f = \sigma \mathbf{E} = \rho_f \mathbf{v} \quad (3)$$

From the important conclusions obtained above,we can list the *Maxwell's Equations* in PLI:

Known that in PLI, $\rho_f = 0, \sigma = 0$.In the linear materials,the constitutive relations show that : $\tilde{\mathbf{D}} = \epsilon \tilde{\mathbf{B}}, \tilde{\mathbf{B}} = \mu \tilde{\mathbf{H}}$ And in the PLI,we express wave functions as followings:

$$\begin{aligned} \tilde{\mathbf{E}}(\mathbf{z}, t) &= \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)} \\ \tilde{\mathbf{H}}(\mathbf{z}, t) &= \tilde{\mathbf{H}}_0 e^{i(kz - \omega t)} \end{aligned}$$

Therefore,we can get the *Maxwell's Equation* as follows:

$$\begin{aligned} \nabla \cdot \tilde{\mathbf{D}} &= \mathbf{0} \quad (\text{Gauss's law for electricity}) \\ \nabla \cdot \tilde{\mathbf{B}} &= \mathbf{0} \quad (\text{Gauss's law for magnetism}) \\ \nabla \times \tilde{\mathbf{E}} &= -\frac{\partial \tilde{\mathbf{B}}}{\partial t} \quad (\text{Faraday's Law}) \\ \nabla \times \tilde{\mathbf{H}} &= \frac{\partial \tilde{\mathbf{D}}}{\partial t} \quad \because \mathbf{J}_f = \mathbf{0} \quad (\text{Maxwell's Law}) \end{aligned}$$

Intrinsic Impedance in PLI

Definition 1 (Intrinsic Impedance η). It shows the ratio of the electric field to the magnetic field in a propagating electromagnetic wave.The physical quantity can provide us *resistance(complex amplitude ratio),the distribution of energy and the phase lag*.

Derivation (Intrinsic Impedance). Begin with Faraday's law and Maxwell's law :

$$\nabla \times \tilde{\mathbf{E}} = -\frac{\partial \tilde{\mathbf{B}}}{\partial t} \quad (4)$$

$$\nabla \times \tilde{\mathbf{H}} = \frac{\partial \tilde{\mathbf{D}}}{\partial t} \quad (5)$$

The wave equation in z direction:

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}$$

$$\tilde{\mathbf{H}}(z, t) = \tilde{\mathbf{H}}_0 e^{i(kz - \omega t)}$$

By the vector calculus formula $\boxed{\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times \nabla f}$ and (4):

$$\nabla \times (\tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}) = -\frac{\partial \tilde{\mathbf{B}}}{\partial t}$$

Expanding the left-hand side (LHS):

$$e^{i(kz - \omega t)} (\nabla \times \tilde{\mathbf{E}}_0) - \tilde{\mathbf{E}}_0 \times \nabla e^{i(kz - \omega t)} = -\frac{\partial \tilde{\mathbf{B}}}{\partial t}$$

Because $\tilde{\mathbf{E}}_0$ is a constant, the LHS first term will be zero and thus the LHS can be reduced to $-\tilde{\mathbf{E}}_0 \times \nabla e^{i(kz - \omega t)}$:

$$-\tilde{\mathbf{E}}_0 \times \nabla e^{i(kz - \omega t)} = -\frac{\partial \tilde{\mathbf{B}}}{\partial t}$$

Here, the wave vector (the direction of wave propagation) $\mathbf{k} = k\hat{k}$ is in the z-direction, that is $\hat{k} = \hat{z}$:

$$\begin{aligned} -\tilde{\mathbf{E}}_0 \times [\hat{z}(ik)e^{i(kz - \omega t)}] &= i\omega\mu\tilde{\mathbf{H}}_0 e^{i(kz - \omega t)} \\ \hat{k} \times \tilde{\mathbf{E}}_0 &= \frac{\omega\mu}{k}\tilde{\mathbf{H}}_0 = \mu v\tilde{\mathbf{H}}_0 \end{aligned} \quad (6)$$

Similarly, by using (5), we can get :

$$\tilde{\mathbf{H}}_0 \times \hat{k} = \frac{k}{\mu\omega} = \frac{1}{\mu v}\tilde{\mathbf{E}}_0 \quad (7)$$

Properties of EM waves in PLI:

In conclusion, by (6), we can find that:

$$\tilde{E}_0 = \frac{\omega\mu}{k}\tilde{H}_0$$

Hence, the intrinsic impedance η is:

$$\eta = \frac{\tilde{E}_0}{\tilde{H}_0} = \mu \frac{\omega}{k} = \mu v \quad (\Omega)$$

Because η is a real number, we can compare their complex phases:

$$\begin{aligned} \because \quad \tilde{E}_0 &= \eta \tilde{H}_0 \\ \implies E_0 e^{i\delta_E} &= \eta H_0 e^{i\delta_H} \\ \therefore \quad \delta_E &= \delta_H \quad (\text{in phase}) \end{aligned}$$

Furthermore, if we consider the polarization of EM wave: (By(6),(7))

$$\begin{aligned}\hat{k} \times \tilde{\mathbf{E}}_0 &= \eta \tilde{\mathbf{H}}_0 \implies \hat{k} \perp \tilde{\mathbf{H}}_0, \tilde{\mathbf{E}}_0 \perp \tilde{\mathbf{H}}_0 \\ \tilde{\mathbf{H}}_0 \times \hat{k} &= \frac{1}{\eta} \tilde{\mathbf{E}}_0 \implies \tilde{\mathbf{H}}_0 \perp \tilde{\mathbf{E}}_0, \hat{k} \perp \tilde{\mathbf{E}}_0\end{aligned}$$

$\therefore \hat{k}, \tilde{\mathbf{E}}_0, \tilde{\mathbf{H}}_0$ are mutually perpendicular.

Conclusion: EM Wave in PLI

In a nutshell, I summarize the results as follows.

1. $\eta = \frac{\mu\omega}{k} = \mu v$
2. $\eta = \mu \frac{1}{\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}}$
3. $\delta_E = \delta_H$ (in phase)
4. The polarization of E , H field and wave vector k are mutual perpendicular.
5. In vacuum, $\eta = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi (\Omega) = 377 (\Omega)$
6. The intrinsic impedance can be regarded as analogous to impedance in circuit theory. The complex amplitude of the electric field represents the driving force, while the complex amplitude of the magnetic field represents the response. The intrinsic impedance describes the relationship between the electric field and magnetic field.

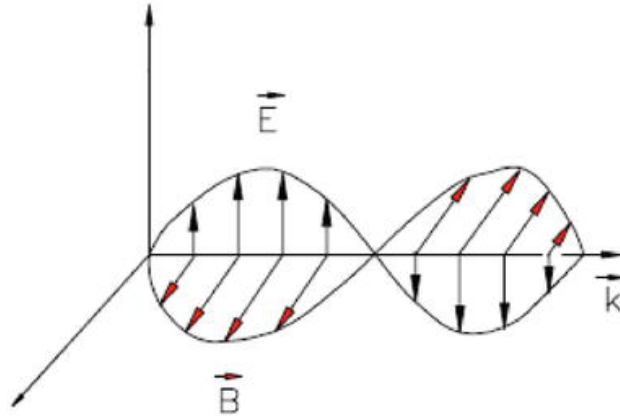


Figure 2: EM Wave

Energy and power for EM wave in PLI:

In fact, there are no source fields inside the PLI but only induction fields.

Definition 2 (Poynting's vector($\mathbf{S}$):). It represents the *energy passing through unit area per unit time* and the direction is same as wave vector $\hat{k}$ (EM Wave propagation direction).

$$\mathbf{S} \equiv \mathbf{E} \times \mathbf{H}$$

In complex form:

$$\tilde{\mathbf{S}} \equiv \tilde{\mathbf{E}} \times \tilde{\mathbf{H}} \text{ (W/m}^2\text{)}$$

$$\begin{aligned}\mathbf{S}(t) &= E_0 \cos(kx - \omega t + \delta_E) \hat{x} \times H_0 \cos(kx - \omega t + \delta_H) \hat{y} \\ &= \hat{z} E_0 H_0 \cos(kx - \omega t + \delta_E) \cos(kx - \omega t + \delta_H)\end{aligned}$$

The definition of the average Poynting vector can be expressed as:

$$\mathbf{S}_{\text{av}} = \frac{1}{T} \int_0^T \mathbf{S}(t) dt$$

By the Trigonometric identity: $\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)]$: and remind that $E_0 = \eta H_0$

$$\mathbf{S}_{\text{av}} = \hat{z} \frac{E_0^2}{\eta T} \int_0^T \frac{1}{2} [\cos(2kz + 2\omega t + \delta_E + \delta_H) + \cos(\delta_E - \delta_H)] dt$$

By the fact that the periodic function integral for a period will be zero, and thus the left side will be reduced to zero.

$$\mathbf{S}_{\text{av}} = \hat{z} \frac{E_0^2}{\eta T} \int_0^T \frac{1}{2} (\cos(\delta_E - \delta_H)) dt$$

Where: $S_{\text{max}} = \frac{E_0^2}{\eta}$, ϕ (power factor) = $\delta_E - \delta_H$:

$$\therefore \mathbf{S}_{\text{av}} = \hat{z} \frac{E_0^2}{2\eta} \cos(\delta_E - \delta_H) = \hat{z} \frac{1}{2} S_{\text{max}} \cos \phi$$

In PLI, $\phi = 0$ (there is no phase lag)

$$\mathbf{S}_{\text{av}} = \hat{z} \frac{1}{2} S_{\text{max}}$$

Discuss the magnitude:

$$I(\text{Average Intensity}) = S_{\text{av}} = \frac{1}{2} S_{\text{max}}$$

In addition, the intensity can connect to u (energy density). *Intensity describes the dynamic flow of energy, whereas energy density represents the energy statically stored in space.* Below, we first review the expression for the energy density of electromagnetic waves.

$$u_E = \frac{1}{2}\epsilon E_0^2, \quad u_H = \frac{1}{2}\mu H_0^2$$

By the relationship of complex amplitude between electric field and magnetic field:

$$u_H = \frac{1}{2}\mu\left(\frac{E_0}{\eta}\right)^2 = \frac{1}{2}\epsilon E_0^2 \quad (\text{Energy is equally distributed in EM wave})$$

$$u = u_E + u_H = \epsilon E_0^2 = \mu H_0^2 \quad (\text{Total energy density in EM Wave})$$

$$u_{av} = \frac{1}{2}u = \frac{1}{2}\epsilon E_0^2 = \mu H_0^2$$

Note 3. We use a more intuitive approach to examine the relationship between the average value and the maximum value. Because Poynting's vector, energy density and power are *cos* function of t . If we take the peak value as the maximum value, by (3) the integral over a period can be expressed by $\frac{1}{2}$ maximum value.

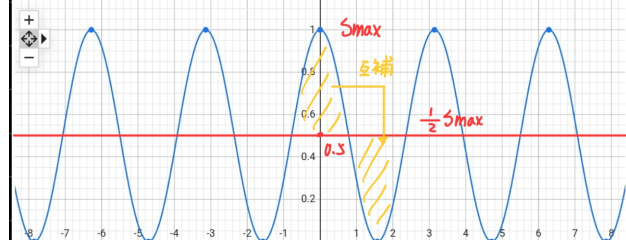


Figure 3: How to realize average value

Therefore, we can summerize as follow:

To begin with,we first talk about power.*Power is energy radiates per unit time.*

$$P = \int \mathbf{S} \cdot d\mathbf{a} \text{ (W)}$$

$$1. S_{av} = I = \frac{1}{2}S_{max}$$

$$2. u_{av} = \frac{1}{2}u_{max}$$

$$3. < u_H > = \frac{1}{2}u_H$$

$$4. < u_E > = \frac{1}{2}u_E$$

$$5. p_{av} = \frac{1}{2}p_{max}$$

Finally, we would like to express it in complex form. The time-average Poynting vector can be expressed in complex form as one-half the real part of the cross product of the complex electric field and the complex conjugate of the magnetic field.

$$\mathbf{S}_{av} = \frac{1}{2}\Re\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$$

$\therefore$ the instantaneous intensity can be expressed as:

$$\begin{aligned} \mathbf{S}(t) &= E_0 \cos(kz - \omega t + \delta_E) \hat{x} \times H_0 \cos(kz - \omega t + \delta_H) \hat{y} \\ \mathbf{S}(t) &= \frac{1}{2}E_0 H_0 \left[\underbrace{\cos(2kz - 2\omega t + \delta_E + \delta_H)}_{\text{time-varying oscillatory terms}} + \underbrace{\cos(\delta_E - \delta_H)}_{\text{constant term}} \right] \hat{z} \end{aligned}$$

the second term is the complex form we get above. Similarly, we can obtain the following result.

$$1. P_{av} = \frac{1}{2}\Re\{\tilde{V} \times \tilde{I}^*\}$$

$$2. u_{av} = \frac{1}{2}\epsilon\Re\{\tilde{E} \times \tilde{E}^*\} = \frac{1}{2}\mu\Re\{\tilde{H} \times \tilde{H}^*\}$$

$$3. S_{av} = u_{av}v$$

Reflection and Transmission on PLI:

When an electromagnetic wave is incident on the interface of two PLI, reflection and refraction occur. Of course, the electromagnetic wave must satisfy both the conditions in the PLI and the boundary conditions at the interface. First, we list several assumptions.

EM wave in PLI:

$\sigma = 0$ and $\rho_f = 0$ with BCs conditions as follows:

$$\tilde{\mathbf{E}}_1^{\parallel} = \tilde{\mathbf{E}}_2^{\parallel}$$

$$\tilde{\mathbf{H}}_1^{\parallel} = \tilde{\mathbf{H}}_2^{\parallel}$$

Types of incidence:

We can sort incidence into two groups:

- Normal Incidence: Wave vector of incident is **normal** to boundary plane.
- Oblique Incidence: Wave vector of incident wave is **oblique** to boundary plane.

1. Normal incidence:

The schematic diagram is shown below.

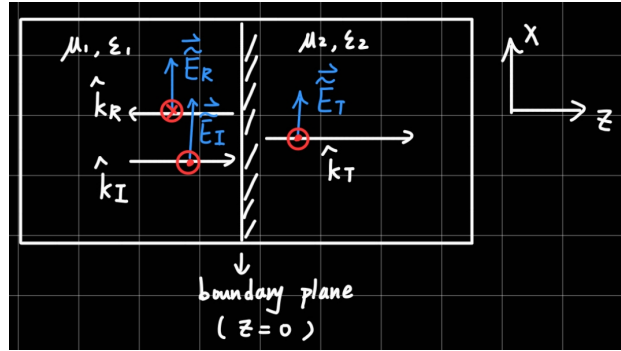


Figure 4: Normal incidence

From figure(4) we find that incident wave propagates in $+z$ direction and we suppose that:

$$\tilde{\mathbf{E}}(z, t) = \tilde{\mathbf{E}}_0 e^{i(kz - \omega t)}$$

$$\tilde{\mathbf{H}}(z, t) = \tilde{\mathbf{H}}_0 e^{i(kz - \omega t)}$$

$$\delta_E = \delta_H$$

Where:

$$\eta = \frac{\tilde{E}_0}{\tilde{H}_0} = \sqrt{\frac{\mu}{\epsilon}} = \mu v = \mu \frac{c}{n} \quad (8)$$

At last, we list three wave functions related to the incidence wave: Assume that the polarization of the electric field is on the x -axis :

$$\begin{aligned}\tilde{\mathbf{E}}_I &= \tilde{\mathbf{E}}_{0I} e^{i(kz - \omega t)} \\ \tilde{\mathbf{E}}_R &= \tilde{\mathbf{E}}_{0R} e^{i(kz - \omega t)} \\ \tilde{\mathbf{E}}_T &= \tilde{\mathbf{E}}_{0T} e^{i(kz - \omega t)}\end{aligned}$$

With the relationship between the electric field and the magnetic field:

$$\begin{aligned}\tilde{\mathbf{H}}_I &= \frac{\tilde{\mathbf{E}}_{0I}}{\eta} e^{i(kz - \omega t)} \\ \tilde{\mathbf{H}}_R &= \frac{\tilde{\mathbf{E}}_{0R}}{\eta_1} e^{i(kz - \omega t)} \\ \tilde{\mathbf{H}}_T &= \frac{\tilde{\mathbf{E}}_{0T}}{\eta_2} e^{i(kz - \omega t)}\end{aligned}$$

where:

$$\begin{aligned}\tilde{\mathbf{E}}_{0I} &= E_{0I} e^{i\delta_I} \hat{x} \\ \tilde{\mathbf{E}}_{0R} &= E_{0R} e^{i\delta_R} \hat{x} \\ \tilde{\mathbf{E}}_{0T} &= E_{0T} e^{i\delta_T} \hat{x} \\ \tilde{\mathbf{H}}_{0I} &= \frac{E_{0I}}{\eta_1} e^{i\delta_I} \hat{y} \\ \tilde{\mathbf{H}}_{0R} &= \frac{E_{0R}}{\eta_1} e^{i\delta_I} (-\hat{y}) \\ \tilde{\mathbf{H}}_{0T} &= \frac{E_{0T}}{\eta_2} e^{i\delta_T} \hat{y}\end{aligned}$$

Note 4. At the interface, $\because z = 0$ $\therefore$ exponent terms will all be the same, that is $e^{-i\omega t}$ and therefore we can cancel the traveling wave terms.

By the boundary conditions (BCs) at the interface:

$$\left[\tilde{E}_{0I} + \tilde{E}_{0R} \right] \hat{x} = \tilde{E}_{0T} \hat{x} \quad (9)$$

$$\frac{\tilde{E}_{0I} - \tilde{E}_{0R}}{\eta_1} = \frac{\tilde{E}_{0T}}{\eta_2} \quad (10)$$

Dropping the unit vector $\hat{x}$ and substituting (9) into (10):

$$\begin{aligned}\frac{\tilde{E}_{0I} - \tilde{E}_{0R}}{\eta_1} &= \frac{\tilde{E}_{0I} + \tilde{E}_{0R}}{\eta_2} \\ \eta_2(\tilde{E}_{0I} - \tilde{E}_{0R}) &= \eta_1(\tilde{E}_{0I} + \tilde{E}_{0R}) \\ \tilde{E}_{0I}(\eta_2 - \eta_1) &= \tilde{E}_{0R}(\eta_2 + \eta_1)\end{aligned}$$

Solving for the reflected amplitude $\tilde{E}_{0R}$:

$$\tilde{E}_{0R} = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) \tilde{E}_{0I} \quad (11)$$

Substituting $\tilde{E}_{0R}$ back into (9) to find the transmitted amplitude $\tilde{E}_{0T}$:

$$\begin{aligned} \tilde{E}_{0T} &= \tilde{E}_{0I} + \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right) \tilde{E}_{0I} \\ \tilde{E}_{0T} &= \left(\frac{2\eta_2}{\eta_2 + \eta_1} \right) \tilde{E}_{0I} \end{aligned} \quad (12)$$

Recall that: $I = \frac{E_0^2}{2\eta}$:

$$R \equiv \frac{I_R}{I_I} = \left(\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \right)^2 \quad (13)$$

$$T \equiv \frac{I_T}{I_I} = \frac{4\eta_2\eta_1}{(\eta_2 + \eta_1)^2} \quad (14)$$

The results also show us the conservation law $\boxed{R + T = 1}$ (The interface does not dissipate energy.) In addition, if we consider the phase shift, we can find that all transmitted wave will **in phase** with incident wave. On the other hand, by (11) the reflected wave may be **out of phase** with incident wave if $\eta_2 < \eta_1$:

Dense material vs. Light material

In nonmagnetic material ($\mu = \mu_0$):

$$\eta = \sqrt{\frac{\mu_0}{\epsilon}} = \mu_0 v = \mu_0 \frac{c}{n}$$

In wave propagation, μ and ϵ together play roles analogous to the “inertia” or “mass density” of the medium.

When an electromagnetic wave enters a material with high μ or high ϵ , the wave must expend additional effort and time to “flip” magnetic dipoles (magnetization) or “distort” electron clouds (polarization). These microscopic resistance and delay mechanisms appear macroscopically as if a mechanical wave were trying to shake a very heavy rope with a large mass density. Consequently, the propagation speed of the wave is reduced.

In conclusion, in PLI material interface: $\boxed{\delta_I = \delta_T \text{ always}}$

$$\delta_R = \delta_I \quad \text{if } \eta_2 > \eta_1$$

$$\delta_R = \delta_I \pm \pi \quad \text{if } \eta_2 < \eta_1 \text{ (phase-inverted reflection)}$$

2. Oblique incidence:

The schematic diagram is shown below.

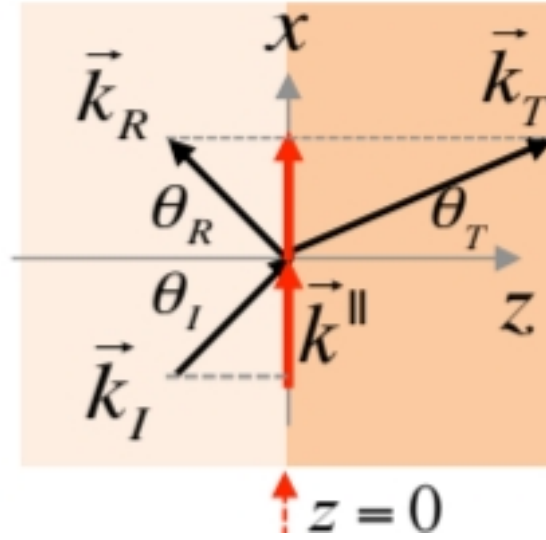


Figure 5: Oblique incidence

By figure 5, we can use BCs on the interface to find the three laws:

- First law: Incidence of plane (three wave vector will lie on same plane (xz - plane))
- Second law (Reflection law) $\theta_I = \theta_R$
- Third law (Refraction law): $n_1 \sin \theta_I = n_2 \sin \theta_T$

We first express that the traveling wave of EM wave as (2D) :

$$e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$$

Where:

$$\mathbf{k} = k_x \hat{x} + k_y \hat{y} + k_z \hat{z}$$

$$\mathbf{r} = x \hat{x} + y \hat{y} + z \hat{z}$$

Note 5. We can now determine the propagation direction of the wave from the inner product of the wave vector and the position vector. That is:

$$\begin{cases} \mathbf{k} \cdot \mathbf{r} < 0 & \Rightarrow \text{(to the left side)} \\ \mathbf{k} \cdot \mathbf{r} > 0 & \Rightarrow \text{(to the right side)} \end{cases}$$

Recall Faraday's BCs:

$$\mathbf{E}_a^\parallel = \mathbf{E}_b^\parallel$$

$$\tilde{\mathbf{E}}_{0I}e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)} + \tilde{\mathbf{E}}_{0R}e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)} = \tilde{\mathbf{E}}_{0T}e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)}$$

Analyze the magnitude of the above expression.

$$\tilde{\mathbf{E}}_{0I} + \tilde{\mathbf{E}}_{0R} = \tilde{\mathbf{E}}_{0T} \quad (15)$$

Analyze the phase of the above expression.

$$e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)} = e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)} = e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)} \quad (16)$$

By (16):

$$\mathbf{k}_I \cdot \mathbf{r} = \mathbf{k}_R \cdot \mathbf{r} = \mathbf{k}_T \cdot \mathbf{r} \quad (17)$$

On the boundary ($z = 0$):

$$\begin{aligned} k_{Ix}x + k_{Iy}y + k_{Iz}0 &= k_{Rx}x + k_{Ry}y + k_{Rz}0 = k_{Tx}x + k_{Ty}y + k_{Tz}0 \\ \implies \boxed{\mathbf{k}_I^\parallel = \mathbf{k}_R^\parallel = \mathbf{k}_T^\parallel} \end{aligned} \quad (18)$$

By (18) and the perpendicular of three wave vectors is on z axis:

$$\left. \begin{aligned} \mathbf{k}_I^\parallel &= \mathbf{k}_R^\parallel = \mathbf{k}_T^\parallel \\ \mathbf{k}_I^\perp &= k_I^\perp \hat{z} \\ \mathbf{k}_R^\perp &= k_R^\perp (-\hat{z}) \\ \mathbf{k}_T^\perp &= k_T^\perp \hat{z} \end{aligned} \right\} \implies (\text{they are coplane on } xz \text{ plane})$$

Besides, (18) and notice Figure 5 show that: $\boxed{k_I \sin \theta_I = k_R \sin \theta_R = k_T \sin \theta_T}$

Derivation (Reflection Law).

$$\begin{aligned} \because k &= \frac{\omega}{v} = \frac{\omega}{c} n \implies k \propto n \\ n_I \sin \theta_I &= n_R \sin \theta_R \\ \because k_I &= k_R \\ \therefore \sin \theta_I &= \sin \theta_R \implies \theta_I = \theta_R \end{aligned}$$

Derivation (Refraction law(Snell's law)).

$$\begin{aligned} \because k_I \sin \theta_I &= k_T \sin \theta_T \\ \therefore n_I \sin \theta_I &= n_T \sin \theta_T \implies n_1 \sin \theta_I = n_2 \sin \theta_T \end{aligned}$$



Figure 6: TIR

Total Internal Reflection(TIR)

When an EM wave propagates from a denser medium into a rarer medium, total internal reflection occurs as the incident angle increases to the critical angle.

1.3.1 Lorentz oscillator model

Consider the EM wave in perfect linear insulator ($\mu, \epsilon, \sigma = 0$). Here, the energy dissipation is caused not by free charges but by bound charges. We consider the Lorentz damping model to explain this phenomenon. Bound charges (ϵ) experience friction during the polarization process, which consequently leads to energy loss.

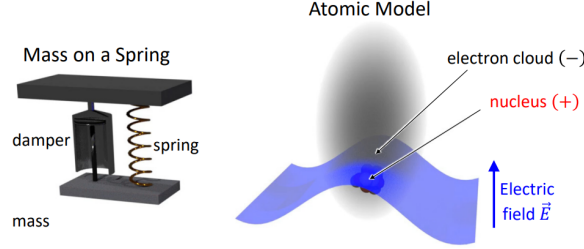


Figure 7: Lorentz oscillator model

By Figure 7, we find it provides an excellent analogy for the oscillation model at the atomic scale. Physicists use the Lorentz model, consisting of a spring-mass system with a damping element, to simulate bound charges in insulators. The small ball represents a negatively charged electron, while the *restoring force originates* from the attraction between the nucleus and the electron. This system possesses a natural oscillation frequency ($\omega_0 = \sqrt{k/m}$). The externally applied oscillating electric field acts as a periodic driving force, causing the electron to undergo forced oscillation. During the interaction between the electron and the external electric field (polarization), the microscopic oscillation of the electron produces friction, which in turn leads to energy dissipation.

Derivation (Polarization of single bound dipole in *Lorentz Oscillator Model*). By analyzing the forces acting on the electron, we can derive the result from Newton's second law.

$$\Sigma \mathbf{F} = \mathbf{F}_{binding} + \mathbf{F}_{damping} + \mathbf{F}_E$$

Where:

$$\begin{aligned} \mathbf{F}_{binding} &= -m\omega_0^2 \tilde{\mathbf{x}} \quad (\text{attraction between nucleus and electron}) \\ \mathbf{F}_{damping} &= -m\gamma \frac{d\tilde{\mathbf{x}}}{dt} \quad (\text{Energy loss}) \\ \mathbf{F}_E &= -q\tilde{\mathbf{E}} \\ \therefore m \frac{d^2 \tilde{\mathbf{x}}}{dt^2} + m\omega_0^2 \tilde{\mathbf{x}} + m\gamma \frac{d\tilde{\mathbf{x}}}{dt} &= -q\tilde{\mathbf{E}} \end{aligned}$$

Assume the solution of $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}_0 e^{-i\omega t}$:

$$\therefore [m(-i\omega)^2 \tilde{\mathbf{x}}_0 + m\omega_0^2 \tilde{\mathbf{x}}_0 + m\gamma(-i\omega) \tilde{\mathbf{x}}_0] e^{-i\omega t} = \tilde{\mathbf{E}}_0 e^{-i\omega t}$$

We eliminate the time-dependent factor on both sides of the equation.

$$\tilde{\mathbf{x}}_0 = \frac{q}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \tilde{\mathbf{E}}_0$$

Given:

$$\tilde{\mathbf{x}} = -\frac{q}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \tilde{\mathbf{E}}$$

We can get the electric dipole moment: ($\tilde{\mathbf{p}} = -q\tilde{\mathbf{x}}$) :

$$\tilde{\mathbf{p}}(\omega) = -q\tilde{\mathbf{x}} = \frac{q^2}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \tilde{\mathbf{E}}$$

By $\boxed{\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}}$:

$$\tilde{\chi}_e(\omega) = \frac{q^2}{m\epsilon_0} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \quad (19)$$

If the molecules within a material possess *different natural angular frequencies* , this indicates that the bound charges are subject to different degrees of binding from chemical bonds or atomic nuclei. As a result, the electric dipoles exhibit different modes. We can express the displacement of j_{th} mode as:

$$\tilde{\mathbf{x}} = -\frac{q}{m} \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \tilde{\mathbf{E}}$$

Note 6. f_j is a weighting coefficient representing the effective number of electrons participating in the oscillation of the j_{th} mode.

$$\Sigma f_j = Z \quad (\text{the total number of electrons in the atom})$$

Therefore, the total microscopic dipole moment in a molecule, composed of different oscillation frequencies, can be expressed as a linear superposition of different modes. That is:

$$\tilde{\mathbf{p}}_{total} = \frac{q^2}{m} \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \tilde{\mathbf{E}}$$

If we consider the total dipole moment per unit volume:

$$\tilde{\mathbf{P}} = N \tilde{\mathbf{p}}_{total} = N \frac{q^2}{m} \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega} \tilde{\mathbf{E}}$$

Where: N is the total number of molecules in material.

Next, we discuss the dipole moment per unit volume for a single resonance mode.

$$\boxed{\tilde{\mathbf{P}} = \frac{Nq^2}{m} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \tilde{\mathbf{E}}} \quad (20)$$

Recall that the constitutive relation :

$$\tilde{\mathbf{D}} = \epsilon_0 \tilde{\mathbf{E}} + \tilde{\mathbf{P}}$$

Where:

$$\tilde{\mathbf{P}} = \epsilon_0 \tilde{\chi}_e \tilde{\mathbf{E}}$$

By (20), we can get the results of complex *Susceptibility* :

$$\begin{aligned}\tilde{\chi}_e(\omega) &= \frac{Nq^2}{m\epsilon_0} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} \\ \Re\{\tilde{\chi}_e\} &= \frac{Nq^2}{m\epsilon_0} \frac{(\omega_0^2 - \omega^2)^2}{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2} \\ \Im\{\tilde{\chi}_e\} &= \frac{Nq^2}{m\epsilon_0} \frac{(\gamma\omega)^2}{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}\end{aligned}$$

In a non-magnetic material ($\mu = \mu_0$) :

$$n(\omega) = \sqrt{\epsilon_r(\omega)}$$

Therefore:

$$\begin{aligned}\tilde{n} &= (1 + \tilde{\chi}_e)^{1/2} \\ \because 1 \ll \chi_e \therefore (1 + \tilde{\chi}_e)^{1/2} &\approx 1 + \frac{1}{2}\tilde{\chi}_e \\ \tilde{n} &= (1 + \frac{1}{2}\Re\{\tilde{\chi}_e\}) + \frac{1}{2}\Im\{\tilde{\chi}_e\}\end{aligned}$$

And we can separate $\tilde{n}$ into real parts and imaginary parts:

$$\tilde{n}(\omega) = n(\omega) + in''(\omega)$$

$$\begin{cases} n(\omega) = 1 + \frac{1}{2}\Re\{\tilde{\chi}_e\} \\ n''(\omega) = \frac{1}{2}\Im\{\tilde{\chi}_e\} \end{cases}$$

We can make a $(n, \alpha) - \omega$ curve to observe property of material in different operating frequency.

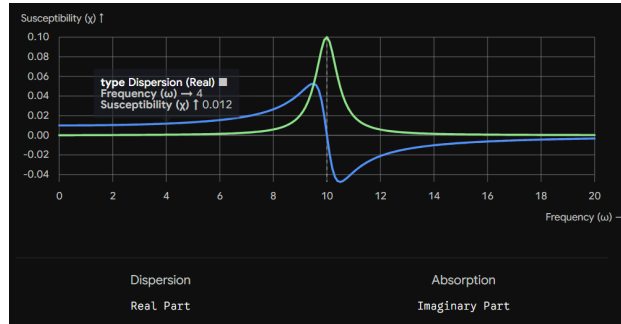


Figure 8: single resonance model

Following resources can be very helpful. [The introduction to Lorentz model](#) & [nice video\(1\)](#) & [nice video\(2\)](#)